

Mechanically Passive Haptic Feedback Device for e-NABLE Phoenix V3 Hand

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Abstract

Extensive research has been focused on adding more functions to prosthetics. Haptic feedback has become a very important function, but most of the devices are limited by high cost, and battery power. This study aimed to create a device that was able to provide force feedback in a non battery powered way. This was achieved using a mechanically passive system. This study presents the design, fabrication, and testing of this device. This device is able to provide variable force feedback through strings. At the best grasping position, a 17 newton (N) input force is able to be transmitted through the device resulting in a 2N output force.

Keywords: Haptic Feedback, 3D printed, Prosthetics, Mechanically Passive, e-NABLE

Mechanically Passive Haptic Feedback Device

3D printed prosthetics have grown in popularity as they have offered a low-cost, personalized, and quick alternative to more traditional prosthetics.^[1,2] 3D printing is a form of manufacturing that involves laying out plastic layer-by-layer. 3D printed prosthetics have been widely used, both body powered, and battery powered systems.^[3,4] These prosthetics offer great benefit, but many of these devices do not incorporate haptic feedback systems. Haptic feedback systems aid in the sense of touch and help patients gain a better understanding of what they are touching.^[5] There have been many haptic feedback systems implemented into 3D printed prosthetics, but almost all of these require battery power. Battery power causes the need for recharging which can become a hassle. The body of knowledge cited the need for more work that focuses on creating a haptic feedback system that can be implemented into a 3D printed prosthetic without the need of battery power. Creating a non-electric device can add to the reliability of the haptic feedback system, and remove the burden of the user having to charge the device.

Prosthetics

Prosthetics are devices that replace missing limbs. They can act as arms or legs to help someone who has lost a limb gain functional mobility. They can also help comfort the user as they would be able to have a replacement for their limb. Over the past few years, prosthetics have increased in function. This is very helpful as many people around the world lose their limbs, which leads to an increase in demand for artificial limbs. Keeping or improving these functions in these prosthetic limbs are a necessity. Upper-Limb prosthetic devices are prosthetic devices including a hand and an arm mechanism. These devices are used to help users interact with objects and can be customized to fit each individual. Body-powered prosthetics are

“considered the standard of care”^[6] due to their simplicity and value. These devices function by using the user's working arms or shoulder to move cables that open and close the prosthetic limb. Many body-powered devices use shoulders or elbows to open and close the prosthetic, this allows the user to precisely control the prosthetic device, and gain a small amount of haptic feedback through the tension in the strings.^[7]

Recent advancements have been made in prosthetics, especially in 3D printing and nerve cell interaction. These recent advancements have created prosthetics that are closer to mimicking a human limb; however, prosthetics still face problems. Creating prosthetic devices at a lower cost, with increased biocompatibility, and improved sensory feedback are still major challenges.^[1] These challenges are being addressed with the help of 3D printing technology; although, they still need improvements. If prosthetics do not meet requirements or functionality for a patient, some choose not to wear them altogether. This is called rejection and is a major issue presenting prosthetic devices as many people feel they are a hassle or are not practical enough to wear.^[8] This presents a major issue, because even if the prosthetic devices are accurate and reliable, if they are uncomfortable or the user doesn't want to wear them, they are rendered useless.

3D printed prosthetics

3D printing is a form of additive manufacturing. This manufacturing technique involves adding material in layers, as opposed to removing it from a block or by molding. This allows manufacturing “complex geometries with high precision and minimal material waste” possible.^[9] 3D printing provides a three dimensional axis for printing objects which allows complicated shapes to be easily printed. This can be very helpful for fast and customized prototyping. Many people find prosthetics custom formed for them more comfortable than a prosthetic made for

mass production. Traditionally, custom prosthetics are made by molding a patient's hand, casting the mold, then creating a plastic modified mold out of the cast. This creates a personalized prosthetic, however, it involves a long and complicated process that can be tedious. 3D printing provides a quick and effective way to personalize prosthetics.^[10]

3D printed prosthetics have grown in popularity as they can be very easily and quickly created, in addition, they are also relatively cheap to manufacture. 3D printed upper-limb prosthetics are usually body powered and consist of five fingers. They are controlled by pulleys in a way that when you extend or contract a body part, like an elbow, the prosthetic hand opens and closes. 3D printed prosthetics are also useful for their ability to be customized. In recent years, companies like e-NABLE, and Mao3D have started offering files that can be printed on ordinary household 3D printers. These companies offer prosthetics, and materials for people who cannot afford the higher end alternatives. This offers great benefits because it allows people from around the world to have access to these prosthetics without needing expensive equipment or specialists to sculpt and cast their limbs. This study presents a novel haptic feedback system for a 3D printed prosthetic from e-NABLE. e-NABLE is a global open source 3D printing prosthetic company that focuses on creating low cost prosthetics.^[11]

Haptic Feedback

Haptic feedback is a type of sensory feedback that has to do with touch. Sensory feedback is a pressing issue due to the fact that prosthetic users can feel little to no stimuli or feedback from their prosthetic devices.^[1] Prosthetic users get little to no feedback when interacting with an object, which can make the objects hard to grasp and manipulate. Most haptic feedback devices are located on a finger of a prosthetic and give the prosthetic user an indication when the finger is pressing on something. Mechanotactile feedback is a type of indication

method in the form of pressure or force.^[12] The force is usually applied by a motor pushing on a patient's skin. Many feedback systems are battery powered and involve a force sensor connected to a motor. When the force sensor is triggered, the motor vibrates or pushes into the skin giving the user a tactile sensation about the amount of force on the finger. It is possible to create a system without battery power. Mechanically passive systems involve air chambers, cables, and springs to help transfer the input pressure into an output force. For both electric and non-electric feedback types, the feedback is usually only applied to the thumb and pointer finger, however, it can be applied to every individual finger to provide more sensory detail.^[4]

Learning how to identify and recognize general feedback from body-powered prosthetic devices without haptic feedback sensors can provide the user with a sense of grip strength and position.^[13] While this can help, the feedback is minimal and does not allow the user to feel pressure on each of their fingers. Being able to feel pressure is necessary for identifying both the shape of an object, and the grasp the prosthetic has on an object.^[5] Implementing haptic feedback into prosthetics can also help reduce the amount of people who chose to reject and not wear their prosthetic, by providing the prosthetic user with a sense of touch great enough to justify the use of wearing a prosthetic.^[8,14] Many expensive prosthetic devices include haptic feedback because they can use some of the power from the prosthetic for sensing equipment, in contrast, most body-powered prosthetic devices are cheaper and do not include haptic feedback. Due to water and dirt being able to damage batteries in prosthetic devices,^[15] Many people voice their concern for the reliability of both the battery and electrodes in a prosthetic device.^[16] If people wear battery powered devices, they may have to remove them near water, or have to take extra measures to make sure their prosthetic can still function and will not get destroyed. This

increases the hassle of the prosthetic, and can increase the likelihood of them rejecting and not wearing it entirely.

Research Gap and Purpose

Although lots of research has been done on haptic feedback, 3D printing, and body powered prosthetics, the research on the interaction between these three topics has been scarce within the body of knowledge. Many studies have looked into how haptic feedback can be implemented into 3D printing prosthetics, but almost all of these studies used battery powered haptic feedback systems. There has also been, while not as common, some research done on body powered or passive haptic feedback systems, but these studies did not focus on creating 3D printable solutions. The gap this paper will address is the implementation of passive haptic feedback systems in 3D printable body powered prosthetics. This raised the question: “To what extent can 3D printed haptic feedback systems be used to simulate human touch in body powered prosthetic devices?” It was hypothesized that 3D printed systems could be used to simulate human touch in body powered prosthetics using a mechanically passive system with strings to transfer an input to an output force.

The overall purpose of this study was to design and implement a haptic feedback system into a body powered prosthetic all while being 3D printed. This study is very relevant as many people are losing limbs. Creating a way to bring back some sense of touch in a cost effective process is very desirable for patients because it would allow people of all economic backgrounds to get a better experience, and also gain greater function while using their prosthetics.

Method

Research Approach

This study measured the accuracy of the device at different forces and angles. These measurements were numerical, therefore, the most appropriate research approach was quantitative. A quantitative approach allowed for a fixed measurement that could be easily interpreted and compared to other available models. Quantitative measurements also removed all bias and created an objective conclusion. Fixed numerical measurements simplified the process to identify the success of the haptic feedback device. This research study used both a technical descriptive method and a small experimental validation component. This approach has been used before,^[17] and has provided a system where new prototypes and systems can be presented and compared to a value to measure success. This is easily interpreted and easily reproducible.

Research Method

A technical descriptive method aligned with this study since it offered a very in depth overview of how the haptic device in this study was created. There are not many available haptic feedback devices of this nature to test against, so this study was mainly made to explain this new system and how it worked. A technical descriptive method was used to explain the prototyping phase. This was very important in showing how the device was improved, and how it started. A technical descriptive method also showed how it was manufactured and provided information on how this system could be recreated and showed what system and print settings were used. This method also provided information on how this system worked mechanically. Providing information on how it worked mechanically helped show how the actual system worked, and helped add information to help troubleshoot whether or not a user is having problems with it.

Overall a technical descriptive method creates an environment where this study could be reproduced by users who want to build haptic feedback devices for themselves.

This study also used an experimental validation element. This validation element determined that the device was able to function accurately, and was vital in providing statistics that were used to help show that this device was worth using. This experimental method also provided useful force measurement that could be compared to other devices. This experimental validation used force sensors to test input and output force. The forces were measured in newtons (N). Input force was force from the finger element, and output force was force exerted to the body. The input and output force relationship showed both the reliability and accuracy of the system.

Sampling

Convenience sampling was used for the prosthetic and printing material selection. Convenience sampling was used to find these parts as the goal of this article was for the device to be printed and created using common, easily accessible resources. The e-NABLE Phoenix V3 hand model was used as it is open source, popular, easy to print and assemble, and it has a strong grip compared to many other devices. PLA plastic was used as it is popular and cheap. These choices allow the device to be constructed in many different areas with different availability to resources. The string selection used convenience sampling, to make the device more accessible. Necklace string was used in this experiment as it was strong, nonelastic, and semi-stiff. Strength was an important attribute of the string as this device was under tension at all times. If the string were to break under this stress, the device would no longer work. Elastic string was not used as it transferred the input force into extending the string, instead of transferring that force to the

output force pin. Having a stiff wire allowed for easy assembly as a large portion of the building process was pushing the string through something.

Validity

This experiment used a Vernier Dual-Range Force Sensor connected to a LabQuest 3 to measure force in newtons. This ensured content validity in the measurement instrument because this study focused on a force feedback system, and a force sensor measures that best. The 3D modeling software Autodesk Fusion 360 also ensures content validity as computer aided design (CAD) models are needed to allow the printer to create parts. The haptic device was tested multiple times in a controlled environment to ensure both internal reliability, and test reliability.

Data Collection

In the building phase, Autodesk Fusion 360 was used to model the device. After the device was modeled, it was printed on the Bambu Labs P2S printer out of PLA Tough+ Bambu lab filament (see Table 1). This was then connected to a low friction tube (PTFE bowden tube). A string was then passed from the finger, through the tube, and to the output pin.

In the experimental validation section, the device was laid out straight and two Vernier force sensors were laid out so that one was touching the input finger, and the other was touching the output pin. The output sensor was clamped to the output pin to ensure the string was tensioned. The finger pin was then pressed into the force sensor until it reached 17N (± 4 N margin of error). The force of both the input finger and the output pin were recorded and graphed during this time. This was done multiple times to ensure reliable results. This procedure was then repeated with various different curves. The graphs were then analyzed.

Table 1: Characteristics of the haptic feedback device

Quantity of Polylactic Acid (PLA)	9.54 grams
Quantity of string	2ft (length of tube +1ft)
Bowden tube	(4mm OD x 1.5mm wall x 1ft)
Infill Density	10%
Layer Height	0.2
Shell Thickness	1mm
Supports	Yes (tree auto)
Finger	Shorter Distil
Cost of Polylactic Acid (5 distal system)*	~(\$0.50)*5
Cost of bowden tube (5 distil system, each tube 1 meter long)*	\$13.99
Cost of string (5 distil system)*	\$2.49 for 30ft
Total cost for 5 distil system*	~\$18.98

* Cost as of 2026, the price may vary over years

Data Analysis

Data was analyzed by comparing input to output force. This measured the efficiency of the system. Peak force was also collected to show the strength needed to fully actuate the device and how sensitive it was. By comparing it at different curves, it shows where the device works best, and where its weaknesses are. This allowed an in depth analysis of how the device functioned, and it also provided data to be compared with different haptic feedback systems.

Results

Purpose

The purpose of this study was to introduce and describe a purely mechanical and easily 3D printable haptic feedback system that can be implemented into an open source e-NABLE prosthetic model. This would allow people who do not have as many resources, or people who are not satisfied with their prosthetics to be able to regain their sensory feeling again and to be able to sense the pressure on their prosthetic fingers. This study used a Labquest system to collect data. Verener Dual-Range Force Sensors were chosen as they have a very high accuracy ($\pm 0.2\text{N}$ margin of error) and because of their ease of use.

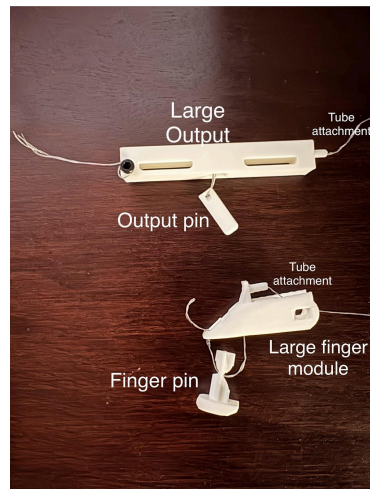
Mechanical Design

The haptic feedback device consisted of an input (located on the finger), and output (located on the arm of a patient). The device transferred the input force from the finger to a pin located on a patient's arm to allow them to detect the forces coming from the finger. This has similar mechanics to bike brakes where tensioning and pulling on the lever tightens the string and tightens the brakes. This system was simple and easy to put together.

Assembly

The device was put together by first tying a strong non-elastic string (Neclase wire was used in this experiment) to the larger finger module. The string was looped around the top of the “+” shaped cutout in the large finger module (See Figure 1). Next, the wire was passed through the small opening in the finger pin. After that, the string was then threaded through the bottom of the larger finger module from the small hole located on the opposite side of the “+” than the tied end. The string was then passed up through the larger finger module and the two finger systems were placed together by lining up the “+”. After that, the wire was then threaded through the tube attachment part of the large finger module. At this point, the tube then was pushed onto the smaller tube attachment portion located on the large finger module. The string was then threaded through the large output part of the device making sure to thread through the tube attachment side. The string was then threaded through the smaller pin portion of the output and passed through the whole output system. After that was done, the tube was then pushed into the tube attachment point, and a screw was threaded through the circular hole located perpendicular to where the string came out of the system. The wire was then threaded around the screw, and the screw was tightened to secure the users preferred tension in the system.

Figure 1: Description and names of the parts of this haptic feedback system



Key Findings

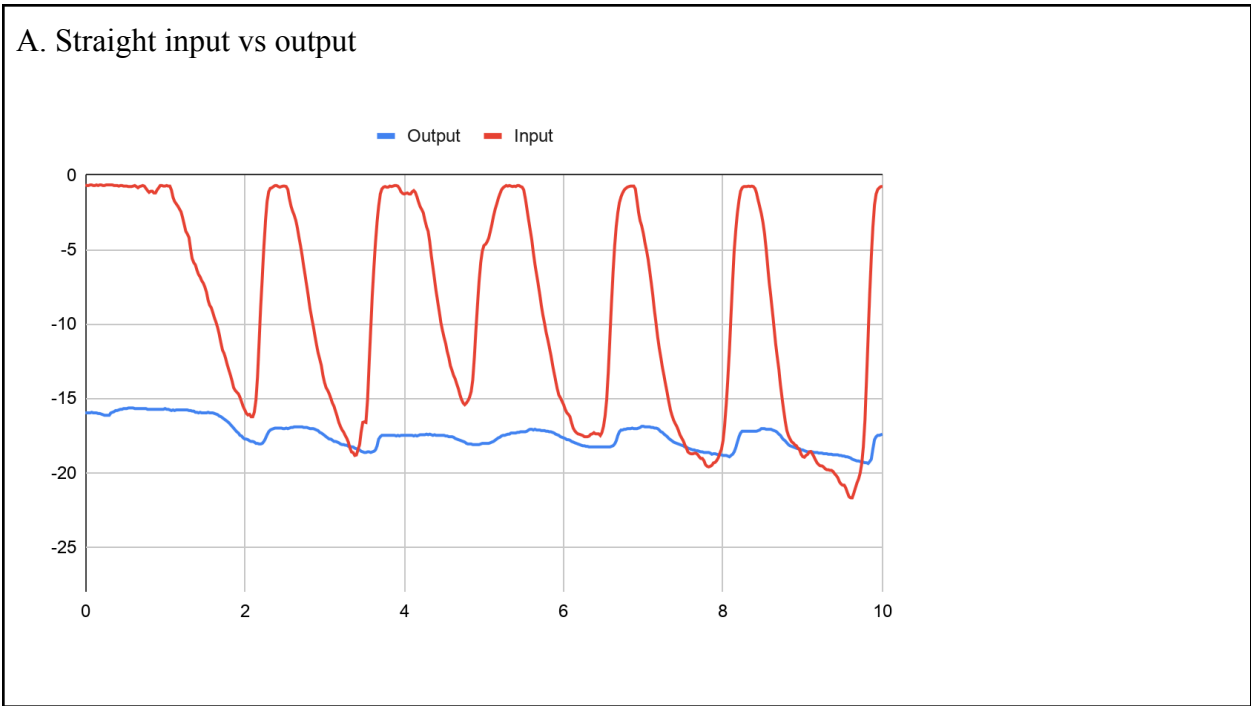
This experiment consisted of laying the haptic feedback device at multiple different angles (See Figure 2). The device was tested at three different positions to make sure the device worked independently of the position of the device. The device was first laid out straight to simulate the prosthetics user when they were not clenching or grabbing anything and in the resting position. The second test (semicurved) was done where the device was slightly bent to simulate the prosthetic user grabbing a large object, or the prosthetic arm semiclosed. The last test (Curved) was done where the device was significantly bent to simulate grasping smaller objects. An input force of $17\text{N}(\pm 2)$ was used for all the tests as it simulated a moderate pressure that prosthetic users would experience on the daily. Through these tests, the device needed to have an output force of above 1N 95% of the time. The device performed well throughout the tests. In all scenarios (30), the device had a 100% success rate with an output of around 2N for each test. This is above the 95% threshold and is deemed a success. Interestingly, for the straight test, the input force caused an output force of $2\text{N}(\pm 1)$ (See Table 2A), the semicurved test had a

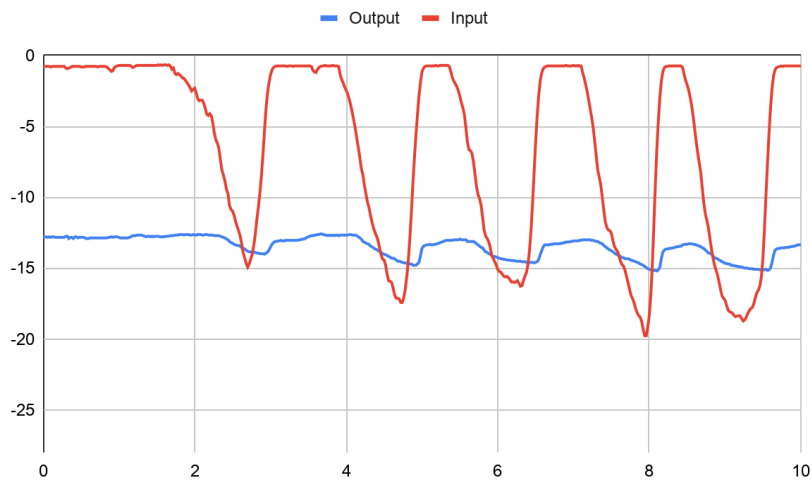
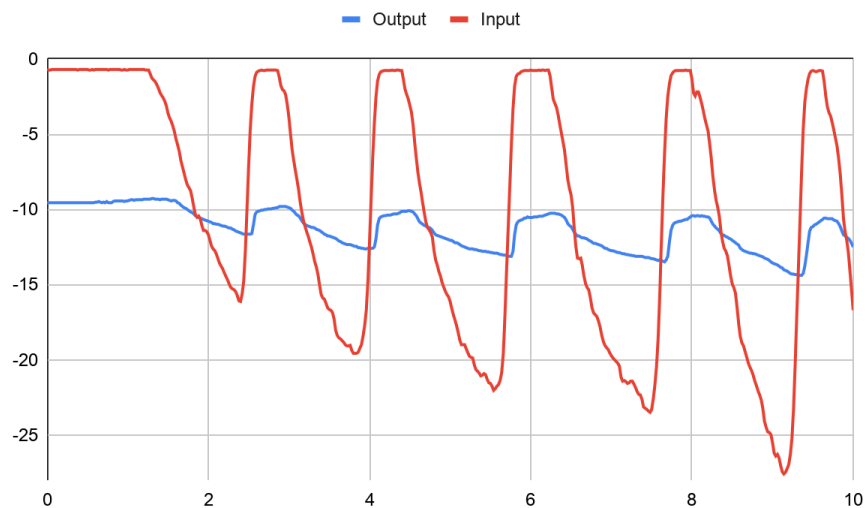
larger output force of 2.5N($\pm$ 0.5) (See Table 2B), and the curved test scored the highest output force at 2.75N($\pm$ 0.5) (See Table 2C). These tests showed that the device worked better in a more grasped position. Overall, a large amount of energy is lost in the process from input to output force, but there is still a noticeable force from the output pin that can be recognized.

Figure 2: Data collection where the input and output were connected to force sensors



Table 2: Input vs output force graphs



B. semicurved input vs output**C. Curved input vs output****Discussion****Research problem**

The issue this study was trying to address was the absence of cost effective, 3D printable, mechanically passive haptic feedback devices. This study created a device that is cheap, 3D

printable, and mechanically passive. The haptic feedback device was able to provide feedback in the form of force. Across different grasping angles, the device met the 95% accuracy rate and was able to show a detectable output force. The device worked most efficiently when it was in the fully grasped, most curved position. At this fully grasped position, and at an input force of $17\text{N}(\pm 2)$, the device was able to output a force of $2.75\text{N}(\pm 0.5)$. The least efficient system was when the device was fully straight, and at the same $17\text{N}(\pm 2)$ had an output force of $2\text{N}(\pm 1)$. The device across all grasping angles provided variable force sensing and pressure. There was a causal relationship between the input and the output force. The device was 3D printed exclusively with PLA filament, used 1 screw, 1 bowden tube, and a 2ft long string.

Implications

This study supports that purely mechanical and 3D printable haptic feedback devices can be created and be functional. This study was supported by a Wurdemann et al. article^[18] that was able to use a purely mechanical system to create haptic feedback. Wurdemann's device used expensive printing equipment, but still helps suggest that battery free systems are viable. The device presented in this study was a battery free haptic feedback device that was made from standard printing filament. This device did include parts that needed to be purchased in stores, but overall this device was more accessible than traditional methods, and was easily created in a household environment with access to a printer and a department store. This device may allow more people to be able to afford this device and therefore cause more people to have access to more functionality in their prosthetics. As the 3D printed e-NABLE Phoenix V3 prosthetics model in this study uses both PLA and long string, the only external resources needed to add this haptic feedback device into the e-NABLE prosthetic hand are screws and bowden tubes. Screws are very accessible and many people have access to them. Bowden tubes are more expensive,

ranging from \$5-20, but they can be replaced with any other, small, inelastic tubing. This causes this system to be very cost effective and easy to replicate. This device is an advancement in functionality for cost effective, 3D printable prosthetics, a field which is rapidly growing.

The experiment showed an input to output graph, and as the input force increased, the output force increased. This shows a relationship between the forces, and was used as an indicator on if the device was successful and there was a relationship. This graph also suggests that this device is able to vary its output force depending on the input force. The data showed that there was a larger output force when the device was in the grasping position. This suggests that the device works better when the device is grasping or when the tube connecting the input to output is most bent. This is most likely due to the bending tube putting tension on the wire inside it.

Similar Studies

This device is significantly less efficient than other battery powered haptic feedback devices, but compared to the most other battery free option, provides a similar output experience. The most common battery free system provides a slightly larger sensor, and uses liquid which provides a more uniform pressure across different grasp distances.^[18] This device did not explicitly mention the input or the output forces. The device used in that study was printed out of TangoPlus filament, a flexible filament which is very expensive and not very common. This makes this device harder to create and less accessible for people. All battery powered prosthetic devices have the ability to change the output force due to the battery being able to provide as much power to the output motor as needed, so these systems are not able to be compared. Overall, there are not many studies to compare to as this is still an emerging field, but from the

existing studies, it shows how battery free devices and low cost devices should continue to be explored.

Alternate Explanation

Some alternative explanations for the results could be where the study was located. In the researchers area, there was access to Amazon, and many other large distribution centers. In many areas, with less development in trade, there is not as much access to less common parts. This can cause prices to go up, and in some cases, for the haptic feedback device to be more expensive and less accessible. This study also heavily relies on the user or creator of this device having a 3D printer. If there is no access to a 3D printer, this device is not able to be made. In this study, broad measurements were used for the different angles of force. Another alternate explanation could be the testing positions. While it was tested in the straight, curved, and semicurved position, these were very broad and didn't follow a specific guideline. These measurements were also isolated from the prosthetic device to increase the internal validity, but that decreased the external validity of not testing on the actual device. This made the conclusions less accurate and less reliable. The input force was also manually pushed by the researcher and not from a specified force actuator. This caused the input forces to not be constant across the different systems, and for the data to be harder to interpret. This could have caused inaccuracies and may have caused some of the output forces to appear larger and therefore more efficient, when in reality, the input force could have just been larger.

Study's Limitations

This haptic feedback device was not able to be tested on an amputee. Being able to test on someone who would actually use the device, and gain their input would allow the device to be tested both quantitatively and qualitatively as opposed to just quantitatively. The device works as

expected and in tests, there was a correlation, but due to human movements and small gaps when attaching the output pin to the body, this device can perform worse, or even not give enough force for the user to recognize. This device also doesn't work well under large loads. In one test, A large input force ($>25\text{N}$) was used, and this caused the string to break in the device, and for the device to become unresponsive. This test was taken out as it had above the 17N threshold and was an outlier, but it shows that the device may not be able to handle huge loads or forces. The output force also does not respond to input stimuli until the input force reaches ($\sim 5\text{N}$) where it would then start to show variation.

Suggestions

This study shows the advantages of having a battery free cost effective haptic feedback device. In future studies, testing this device with human participants and gaining qualitative data would provide further insight into this device and how well it functions. Rejection rate is a huge issue in prosthetic devices as if people do not find it comfortable enough, they will not use it, so being able to limit that rejection rate and make the device as comfortable as possible would be desirable. Future research should also focus on adding more features and abilities to cost effective prosthetic devices. Creating devices that add functionality to low cost prosthetics would be very helpful for people who cannot afford expensive devices and would help shift the prosthetic industry to try to meet everyone's needs.

Conclusion

In conclusion, This study was trying to address the gap in research on 3D printed mechanically passive haptic feedback devices in e-NABLE prosthetic hands. This study was able to design a device that works and provides mechanically passive force feedback. This will be able to help many people who do not have access to large power supplies or who do not like

battery powered devices. This device will also help low income households be able to afford devices that can offer more functionality. A battery free haptic feedback device was created for (~\$18.98), and has a reliable input to output relationship. This device was able to work at different grasping angles with above a 95% success rate and is able to be integrated into the e-NABLE Phoenix V3 prosthetic device.

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